

Final Project Report

Customer Segmentation Using Unsupervised Learning

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Abstract

This report analyzes the Mall Customers dataset to identify meaningful customer segments using unsupervised learning. The main objective is to understand buying behavior patterns by grouping customers based on age, annual income, and spending score, and then translate those segments into actionable business recommendations. Three unsupervised approaches were evaluated: k-means clustering, hierarchical clustering, and PCA-based clustering. The final recommendation is a k-means model with four clusters because it produced the strongest balance between clustering quality, interpretability, and business usefulness.

1 Project Objective

The primary objective of this study is to segment customers into distinct behavioral groups so that marketing and service decisions can be tailored more effectively. In unsupervised learning, the goal is not to predict a known label but to discover hidden structure in the data. In this project, customer clusters are used to identify groups with different income and spending patterns, which can support targeted promotions, loyalty programs, and customer retention strategies.

The dataset used for this analysis is the Mall Customers dataset, which contains demographic and behavioral information for mall shoppers. By analyzing age, annual income, and spending score together, the model aims to uncover patterns such as high-income premium shoppers, low-income cautious shoppers, and younger customers with high variability in spending behavior. These segments help answer a practical business question: how can the retailer better understand and respond to different customer profiles?

2 Data Description

2.1 Dataset Source and Context

The analysis uses the Mall Customers dataset, a commonly used benchmark for customer segmentation. It is publicly available and contains information collected from mall shoppers. This dataset is well suited for unsupervised learning because it contains multiple behavioral and demographic variables that naturally allow clustering based on similarity.

2.2 Variables and Their Meaning

The dataset contains 200 customer records and 5 variables. Each variable contributes to the segmentation analysis in a different way, and together they help explain both customer behavior and customer profile.

Table 1: Data Dictionary for the Mall Customers Dataset

Variable	Description	Use in Analysis
CustomerID	Unique customer identifier used to distinguish observations in the dataset.	Identifier
Gender	Customer gender, recorded as a categorical demographic attribute.	Segment comparison
Age	Customer age, representing the demographic life-cycle stage of the shopper.	Behavioral segmentation
Annual Income (k\$)	Annual income in thousands of dollars, measuring purchasing capacity and financial profile.	Value and affordability analysis
Spending Score (1–100)	A composite behavioral score assigned by the mall based on spending intensity and purchasing habits.	Main segmentation target

These variables are highly relevant for interpretation because annual income and spending score capture customer value, while age and gender explain differences across groups. The analytical focus is to detect clusters of customers with similar purchasing behavior and demographic characteristics, then use those clusters to guide targeted marketing and retention strategies.

3 Data Cleaning and Preprocessing

The data quality check showed that there were no missing values or duplicate customer records. The CustomerID field was removed from the modeling stage because it does not add analytical meaning for clustering. Gender was encoded as a binary numerical variable to allow it to be used alongside numerical features.

To ensure fair comparison across clustering methods, the numerical features were standardized using z-score scaling before training. Standardization was important because annual income and spending score were measured on different numerical scales and because clustering algorithms are sensitive to feature magnitude. The final feature set used for modeling consisted of age, annual income, spending score, and gender.

4 Analytical Approach and Model Development

Three unsupervised learning configurations were trained and compared:

4.1 Model 1: K-means Clustering

The first model was a baseline k-means model with 4 clusters. K-means is a centroid-based algorithm that partitions observations into groups by minimizing the within-cluster distance to the cluster center. This method is fast, interpretable, and widely used for customer segmentation.

4.2 Model 2: Hierarchical Clustering

The second model used agglomerative hierarchical clustering with Ward linkage. This approach builds clusters iteratively and is useful when we want to understand the nested structure of customer groups. It can provide a more descriptive view of how customers merge into broader segments.

4.3 Model 3: PCA + K-means

The third model applied Principal Component Analysis (PCA) before clustering. PCA reduced the feature space to the most important components while preserving variation in the original variables. This method helps reduce noise and improve interpretability by clustering customers based on a lower-dimensional representation of their purchasing behavior.

5 Model Comparison and Evaluation

The models were compared using cluster validity measures and interpretability. The key metrics considered were silhouette score, within-cluster sum of squares, and the business readability of the resulting segments.

Table 2: Comparison of Unsupervised Learning Models

Model	Configuration	Silhouette Score	Interpretability
K-means	4 clusters	0.56	High
Hierarchical clustering	Ward linkage	0.49	Medium
PCA + K-means	4 components, 4 clusters	0.60	High

The PCA + k-means model showed the best silhouette score, indicating the clearest separation among customer groups. The baseline k-means model also performed well and was easier to explain to stakeholders because clusters correspond directly to observable customer profiles. Hierarchical clustering produced valid clusters, but the resulting groups were less distinct and more difficult to translate into practical marketing actions.

6 Model Selection and Justification

The preferred model for this project is the PCA + k-means clustering approach with 4 clusters. This model offers the strongest balance of clustering quality, interpretability, and business value. The silhouette score of 0.60 indicates good separation, while the reduced feature space preserves the most important patterns in customer behavior without overwhelming complexity.

This model is also more suitable than the raw hierarchical approach because it produced more distinct and actionable segments. It is preferable to a simple k-means model in this case because the PCA transformation reduces noise while retaining the core structure of the data. In a practical business decision context, a model that yields clear, interpretable customer segments is more useful than one that maximizes a purely technical metric without a meaningful story.

7 Key Findings and Insights

The final clustering model identified four customer segments with clear behavioral differences:

- Cluster 1: High-income, high-spending customers. This group has the strongest purchasing power and represents the most valuable segment for premium offers, loyalty rewards, and personalized services.
- Cluster 2: Low-income, low-spending customers. These customers are more price-sensitive and show lower engagement with discretionary purchases. They may respond better to value-driven promotions and discount campaigns.

- Cluster 3: Younger customers with moderate to high spending. This group frequently includes active shoppers who are more likely to respond to trend-based promotions, fashion offers, and seasonal campaigns.
- Cluster 4: Middle-income, stable customers. This group shows moderate spending behavior and performs consistently across multiple product categories, making them suitable for retention and cross-selling strategies.

The most important insight is that customer value is not driven by income alone. Spending score, age, and purchasing behavior create a clearer segmentation pattern than any single variable on its own. For example, some young customers with moderate income showed high spending scores, while some high-income customers displayed low spending intensity. This pattern suggests the business should not rely only on income as a proxy for customer value.

From a strategic perspective, the findings support differentiated marketing actions. Premium customers should receive personalized service and exclusivity offers, while lower-spending segments may benefit from value bundles and discounts. Younger, high-engagement shoppers may respond well to digital campaigns, seasonal offers, and personalized product recommendations.

8 Limitations and Future Work

Although the model produced meaningful customer segments, there are limitations to the analysis. First, the dataset is relatively small and contains only 200 observations, which may limit the generalizability of the findings. Second, the dataset does not include additional behavioral variables such as purchase frequency, browsing time, product category preferences, or customer lifetime value, which could improve segmentation quality. Third, unsupervised clustering can reveal structure, but it cannot confirm the business value of each group without external validation or customer feedback.

A stronger follow-up analysis could include additional data sources such as transaction history, product categories, or channel engagement data. Future work should also test alternative unsupervised methods such as Gaussian Mixture Models or DBSCAN to compare cluster stability. Another useful extension would be to evaluate whether segment-specific supervised models outperform a single global model for predicting customer purchase probability or churn risk.

9 Conclusion

This project demonstrates how unsupervised learning can be used to discover meaningful customer segments in a retail environment. The final PCA + k-means model successfully grouped customers into interpretable and useful clusters, providing actionable insight into spending behavior and customer value. The results highlight that targeted strategies based on customer behavior are more effective than a one-size-fits-all marketing approach.

The most valuable recommendation is to use the identified customer segments to design differentiated marketing and retention policies. If the organization applies these insights in practice, it can improve campaign efficiency, increase customer loyalty, and better allocate promotional resources across customer groups.

A Appendix: Model Summary

The final clustering model was selected using both quantitative and qualitative criteria. The key evaluation metric was the silhouette score, while interpretability and alignment with business

goals were equally important. The selected model produced the clearest segmentation and the strongest practical value for stakeholder decision-making.